

The Great British Bake Off

By the Numbers

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Introduction

In this statistical report, we will analyze a dataset from The Great British Bake Off (GBBO). In addition to providing an overview of the observations and variables, it will highlight the winners, the content's age, gender, and race, and any missing data. The central questions explored in this report concern who competes and who wins, particularly with respect to age, gender expression, and apparent race. By examining distributions and frequencies across these variables, this report aims to illuminate patterns that may reflect broader social dynamics within the show. The dataset includes 192 contestants across 16 series, with variables ranging from demographic information to competition performance.

The Data Set

As stated in the introduction, this data set is derived from the GBBO, but let's dive deeper into the variable types and what we can glean from them. The nominal variables in the data are Gender Expression, Apparent Race, Occupation, Hometown, Series, and Publisher. There are only a handful of interval and ratio variables: Age, Number of Star Bakers, and Ranking. Lastly, we have only one ordinal variable in the data set: Placement. With that in mind, the following charts and visuals will show when different statistical measures are appropriate, how to read distributions for both numerical and categorical data, and why it is important to let the data guide our analysis rather than relying on assumptions.

Who Are Our Winners

Referring to Figure 1, we can see the winners of each series, their gender expression, age, occupation, the number of times they won Star Baker, and their approximate hometown. So, let's break it down: out of all the bakers, nine out of 16 are women, meaning seven out of 16 are men, or 56.25% & 43.75%, respectively. As for the winners' ages, we have a minimum of 20, a mean of 32.50, a median of 31, a mode of 30, and a maximum of 60 (an age range of 40). Moving on to Star Baker, we have a minimum of 0, a mean of 1.88, a median of 2, a mode of 2, and a maximum of 5. David Atherton of Series ten won with zero Star Bakers, the only contestant to do so (Edd from Series one is marked as zero Star Baker because the title was not given out during that series). Lastly, the winners' locations are scattered around central England, with one each in Wales and Scotland.

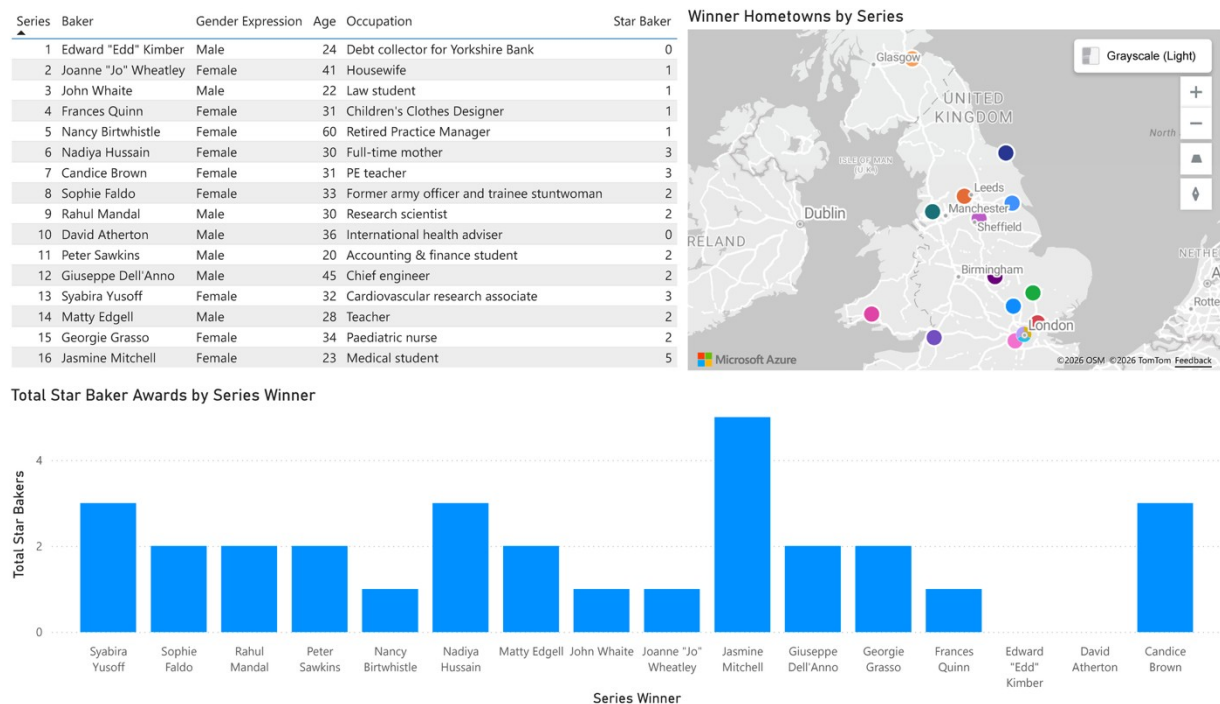


Figure 1

Gender and Race

Breaking down Figure 2 of the contestants and winners, we can see that 9 winners were presented as women, and 97 contestants were presented as women; 13 winners were presented as white, whereas 140 contestants were presented as white. Drilling down into this, we can see that seven of the 16 winners are white women, and 68 of the contestants were white women. This yields 43.75% and 35.42%, respectively, an 8.33-percentage-point gap, indicating that white women win the show at a higher-than-expected rate. It's worth noting that the number of white contestants is declining slightly in later series.

There are several better ways to account for racial identities assigned to a single race. First, and the best in this case, is to add *Multiracial* or *Mixed* categories, since this data is based on observed rather than reported data. Adding one of these categories allows for the acceptance of multiracial identities without assigning them to the contestant. Beyond adding the *Multiracial* or *Mixed* categories, the data set could allow multiple racial identities for each contestant. The best option would be to use self-identification, but because not all contestants do, this is not possible with this data.

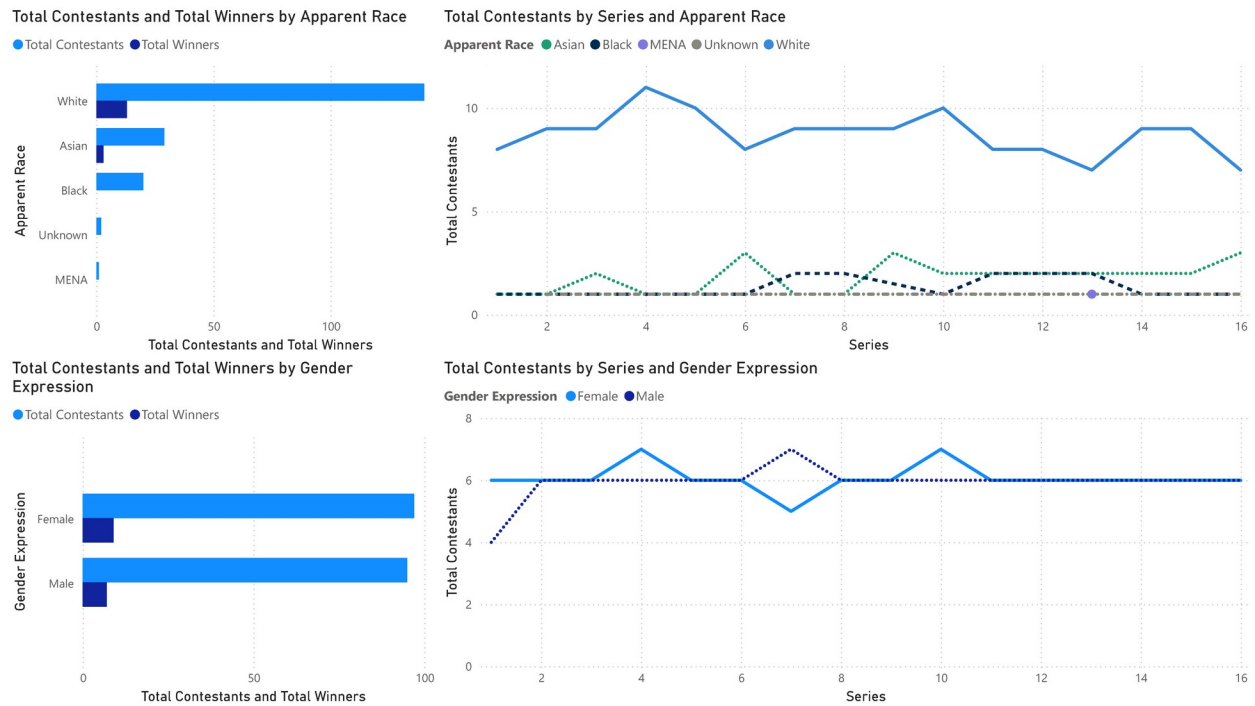


Figure 2

How old?

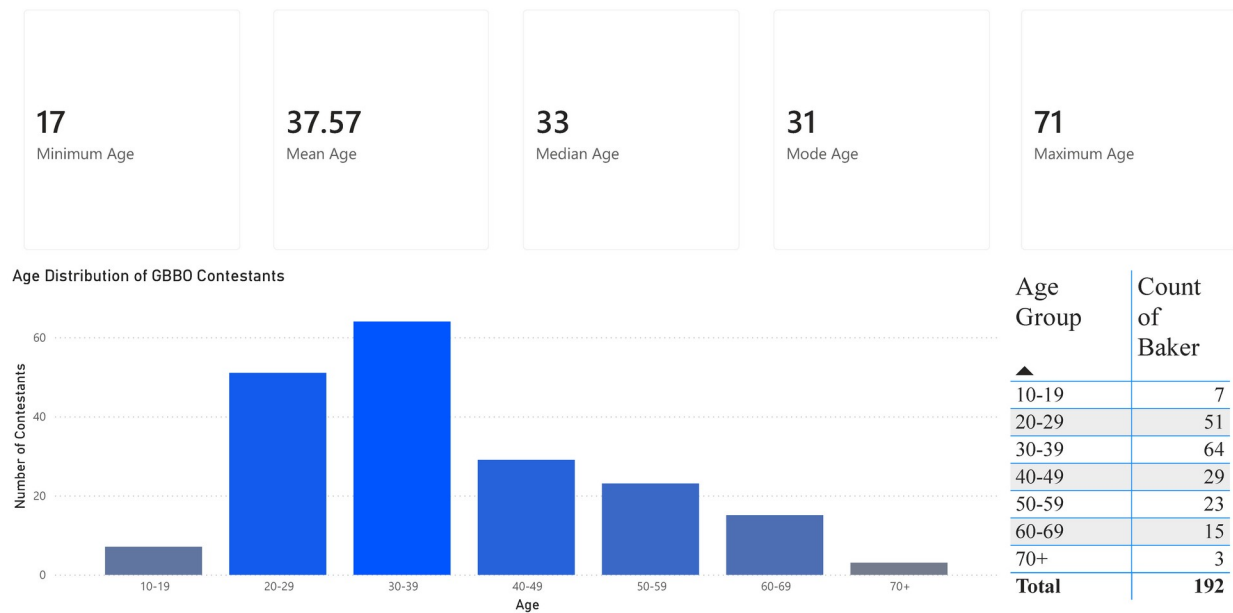


Figure 3

Examining Figure 3, we can see the range of the contestants is 17 – 71 or 54 years; we have a mean of 37.57, a median of 33, and a mode of 31. Looking at our histogram, we can see the distribution is right-skewed, with most contestants between 30-39 years of age, followed by those between 20-29 years of age.

What's missing?

The main variables missing from this dataset are as follows: whether the contestants had any formal culinary training or were entirely self-taught, detailed scoring information, and the contestants' education level, which would help round out the dataset. In addition to expanding this data, a complementary dataset on the judges could help identify any bias they may have.

Conclusion

This descriptive analysis of The Great British Bake Off identifies several notable patterns. Winners are generally younger than the overall contestant pool, with a mean winning age of 32.50 compared to 37.57 for all contestants. White women are overrepresented among winners relative to their proportion in the contestant pool, winning at a rate 8.33 percentage points higher than expected. The age distribution of contestants is right-skewed, with a peak in the 30–39 range. These findings prompt further inquiry into which forms of culinary creativity are recognized and rewarded. The primary limitations of this dataset include reliance on observer-assigned race and gender, absence of culinary training and education data, and lack of detailed episode-level scoring.

Work Cited

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